

A decorative graphic on the left side of the slide consisting of two overlapping parallelograms. The front one is blue and the back one is a light green color. They are positioned diagonally, with the blue one partially covering the green one.

Static Electricity & Magnetism

Grade Eight Science



Key Concepts: Relationships, Transformations

Static Electricity and Magnetism

Statement of Inquiry:

Electrical and magnetic forces fill space as fields; understanding their form and relationships allows us to transform energy in useful ways



Factual and Conceptual Questions

- What is magnetism and what causes it?
- What is static electricity and what causes it?
- What are magnetic and electric fields?
- Why does heating up a bar magnet reduce its magnetic strength?
- What happens when you cut a bar magnet in half?
- Why does hitting a bar magnet with a hammer reduce its magnetic strength?



Pre-knowledge:

What do you know about forces and atoms?

FORCES:

- A force can be defined in simple terms as a push or a pull
- It takes a force to overcome inertia and make stuff move
- <https://www.youtube.com/watch?v=GmIMV7bA0TM>

ATOMS:

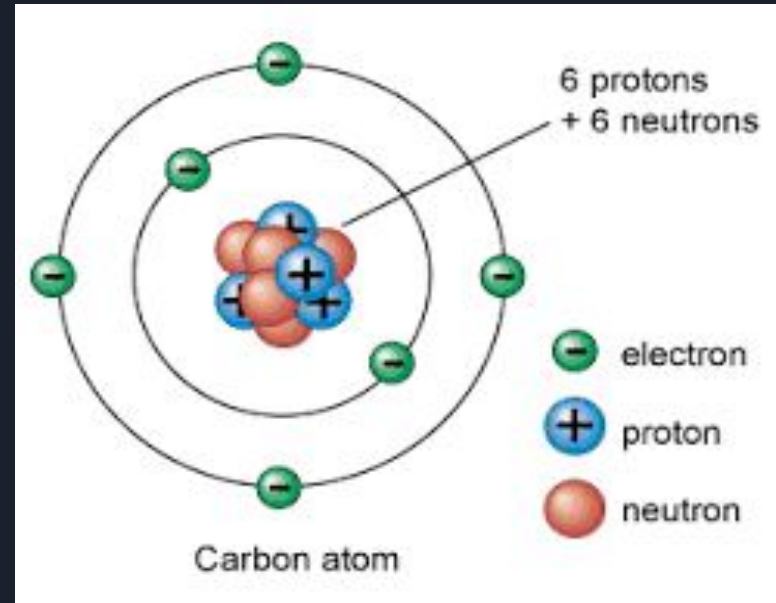
- The smallest unit of matter; all matter is made of atoms
- Atoms are made up of subatomic particles: protons, electrons, and neutrons

Atoms and Subatomic Particles

An **atom** is the smallest unit of matter

Everything is made up of tiny units called atoms

Just like people are made up of blood, bones, tissues, proteins etc, atoms are made up of **subatomic particles: Protons, neutrons, and electrons**





Subatomic Particles

Protons are positively charged particles that live inside the nucleus

Neutrons are neutral particles; they have no charge and live inside the nucleus

Electrons are negatively charged particles that spin in orbit around the nucleus

Protons and neutrons cannot move outside their system because they are bound inside a nuclear membrane

ONLY ELECTRONS CAN MOVE because they are outside the nucleus

- <https://www.youtube.com/watch?v=IP57gEWcisY>



What are Magnets?

- **Magnets** occur where groups of atoms in certain metals, such as iron, cobalt, and nickel, **align in a specific way** - **most of the electrons spin in the same direction**
- This makes the atoms in these substances strongly **magnetic**
- All magnets are dipoles meaning that they have two opposite poles (North and South)

LAW OF MAGNETIC FORCES:

- Like poles repel (south and south or north and north)
- Opposite poles attract (south and north)

What are Magnetic Fields?

- A **magnet** is any material that produces a **magnetic field** around it
- **Magnetic Fields** - the area around a magnet that has magnetic force; **fields** are caused by moving electric charges
- **Magnetic** objects become magnets when other magnetic objects enter their **field**



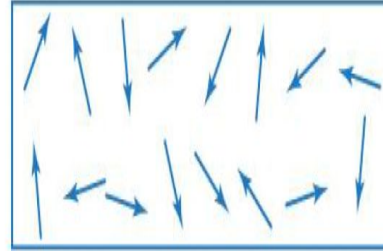


What is Magnetism?

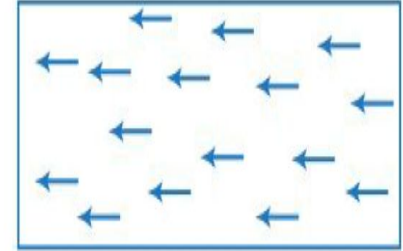
- **Magnetism** is the **force** exerted by magnets when they **attract** or **repel** each other
- **Magnetism** is caused by the motion of electric charges spinning in the same direction
- **Magnetism** is observed by the attraction and repulsion at the poles
- **Opposite Poles attract and Like Poles repel**

What Makes Stuff Magnetic?

- A magnetic material is made up of tiny magnetic areas of spinning electrons called **domains**
- When **domains** are randomly arranged (opposite spins) the magnetic fields cancel each other out - **not magnetic**
- When **domains** (electron spins) are aligned their individual magnetic fields add together making a magnetic field - **magnetic**



▲ In this magnetic substance, all the domains are misaligned. As a result, the individual magnetic fields each act against each other and so there is no overall magnetic field



▲ In this magnetic substance, the domains have all been aligned. The magnetic fields all add together to create a magnetic field

What is Static Electricity?

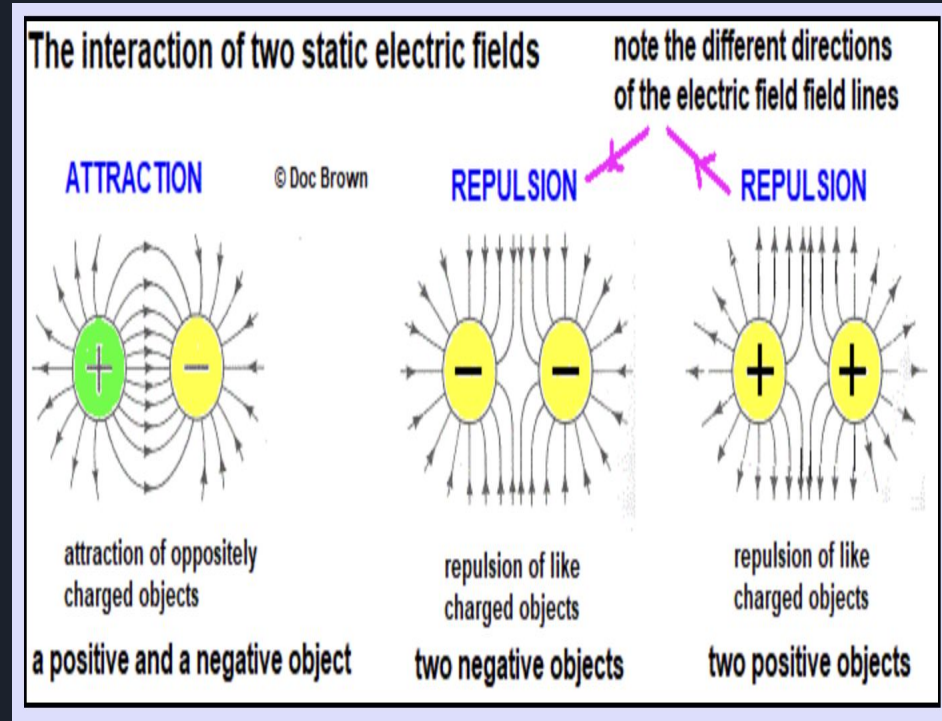
- **Static electricity** is a force exerted by the buildup of charge on the outside of an object
- **Static electricity** is caused by an imbalance of electric charges that create stationary **electric fields**
- **Opposite Charges attract** each other and **Like Charges repel** each other



Source www.twinkl.ca

Electric Fields

- **Electric Field**- the area around a charged object where other charged objects feel a **force**
- **Electric fields** are created around particles or objects that have overall positive (caused by protons) or negative (caused by electrons) **electric charges**
- **Electrons are negative; protons are positive**





Charges (Positive, Negative, Neutral)

Opposite charges attract one another (negative and positive)

Like charges repel each other (two negatives or two positives)

Neutral means that the object has an equal number of positive protons and negative electrons - **neutral objects have no charge**

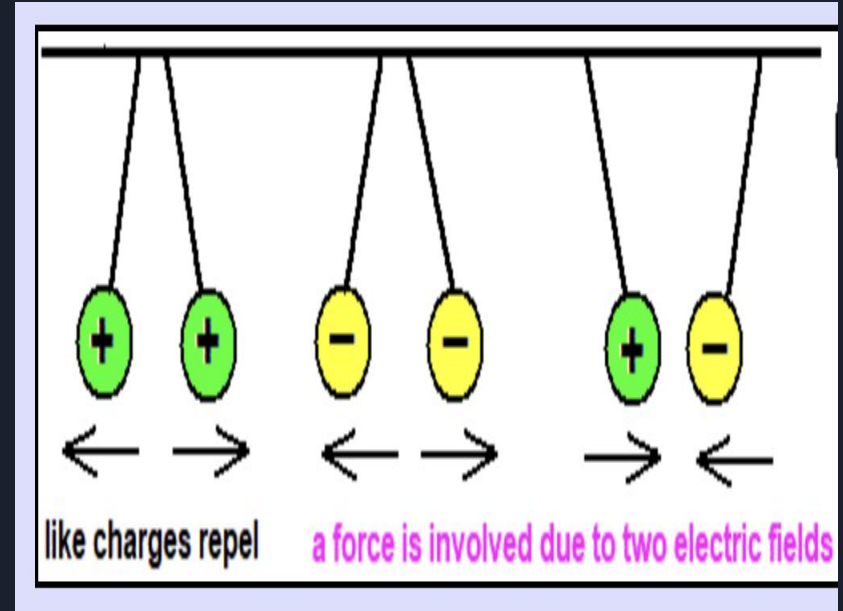
Negatively charged means that an object has more negative electrons than positive protons (imbalance of charge)

Positively charged means that an object has more positive protons than negative electrons (imbalance of charge)

Charges are conserved. Charges can be transferred between objects but they always remain; they are not created or destroyed.

Opposite Charges Attract; Like Charges Repel

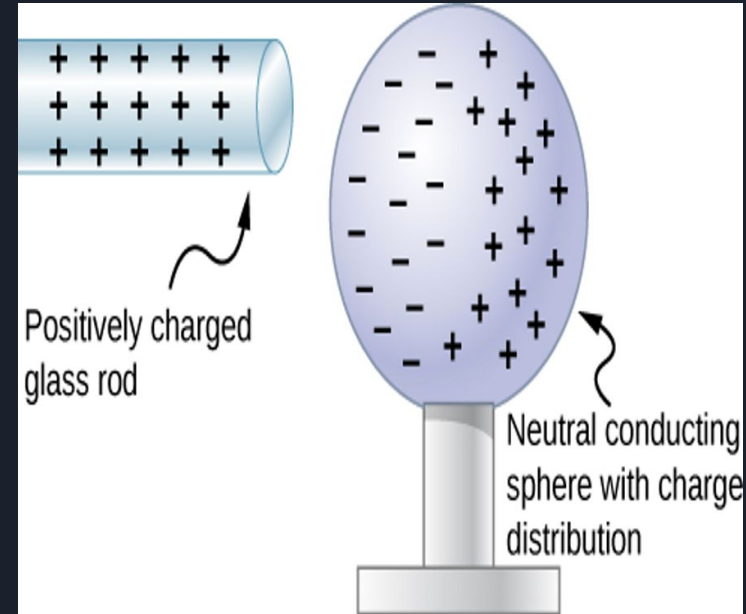
- A negative charge pushes other negative charges away and attracts positive charges
- A positive charge pushes other positive charges away and attracts negative charges



Polarization of Charge

Polarization of charge means that the charges on an object separate into groups of positives and negatives

Polarization of charge occurs when a charged object is brought near a neutral object



Source:

<https://pressbooks.online.ucf.edu/osuniversityphysics2/chapter/conductors-insulators-and-charging-by-induction/>

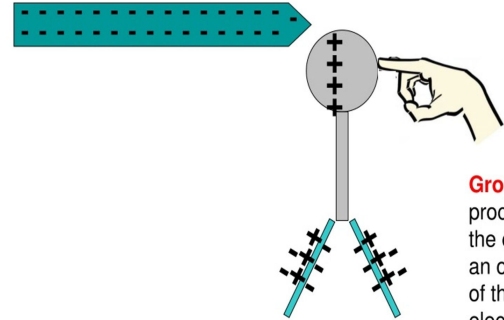
Grounding

Grounding is the process of removing excess electrons from a system in order to bring it back to neutral

A charged object can be **grounded** by coming into contact with a neutral object that it can transfer excess electrons to; The goal of every charged object is to get back to neutral

For example, if you place a neutral hand on the ball of a charged electroscope electrons will transfer from the electroscope to the hand allowing the electroscope to get back to neutral

5. The electroscope is grounded:



Grounding is the process of *removing* the excess charge on an object by means of the transfer of electrons between it and another object of substantial size.

The Lightning Rod: A tool for grounding lightning

The lightning rod was invented by Benjamin Franklin

It is a tall, metal rod that is attached to a house or other building; the rod is taller than the building

Lightning will hit the metal rod (a **conductor**) and travel into the ground rather than hitting the house



Source:

<https://www.kwtx.com/content/news/Lorena--Lightning-rods-installed-on-lightning-damaged-house-378186271.html>



Electron Affinity (EA) & The Triboelectric Series

Electron affinity is the ability of a material to hold onto its electrons

Materials with a **high EA** hold tightly to their electrons

Materials with a **low EA** lose electrons more easily

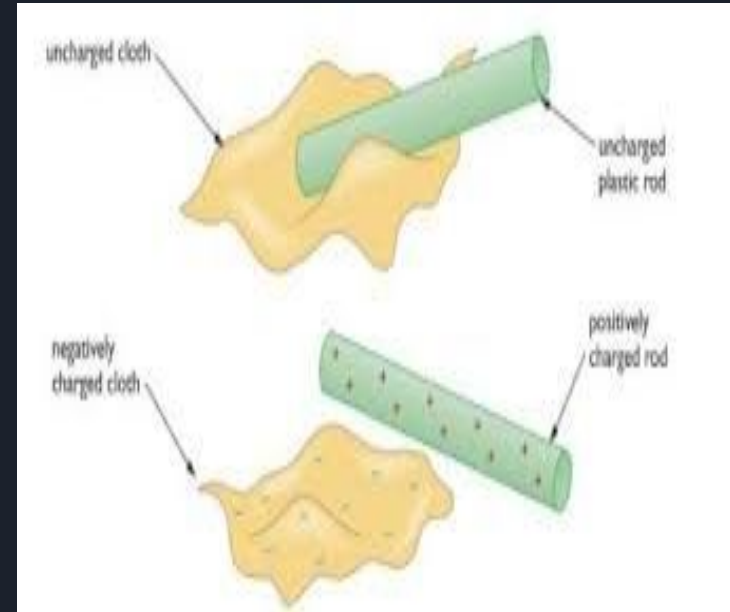
The Triboelectric Series gives a list of materials from high to low EA

Materials that are lower on the list give electrons to materials that are higher on the list as long as the materials are in contact with one another

Types of Static Charging: Type 1 - Friction

Friction occurs when two materials are rubbed together

During friction electrons can move from a low EA object to a higher EA object because the objects are in contact with each other

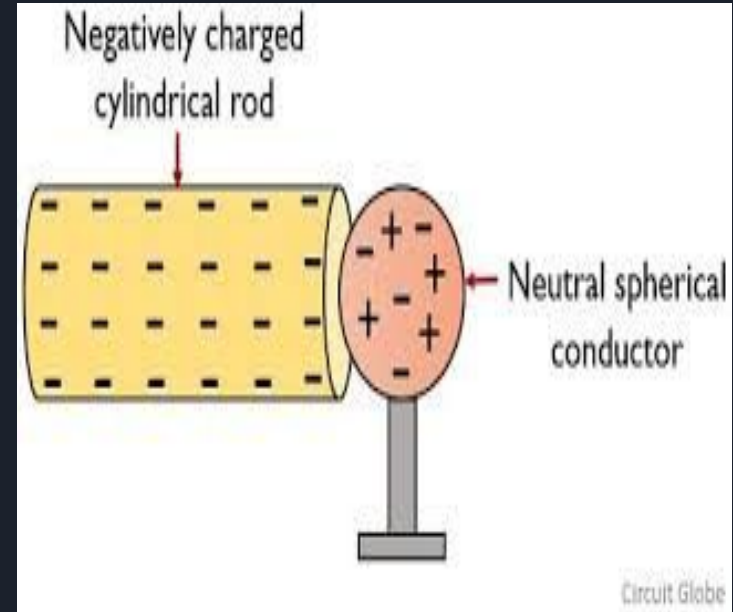


Types of Static Charging: Type 2 - Conduction

Conduction occurs when a charged object, such as a rod, is touched against a neutral object, such as an electroscope

THE GOAL OF EVERY CHARGED OBJECT IS TO GET BACK TO NEUTRAL

So, the rod transfers electrons to the electroscope in order to get back to neutral



Source:

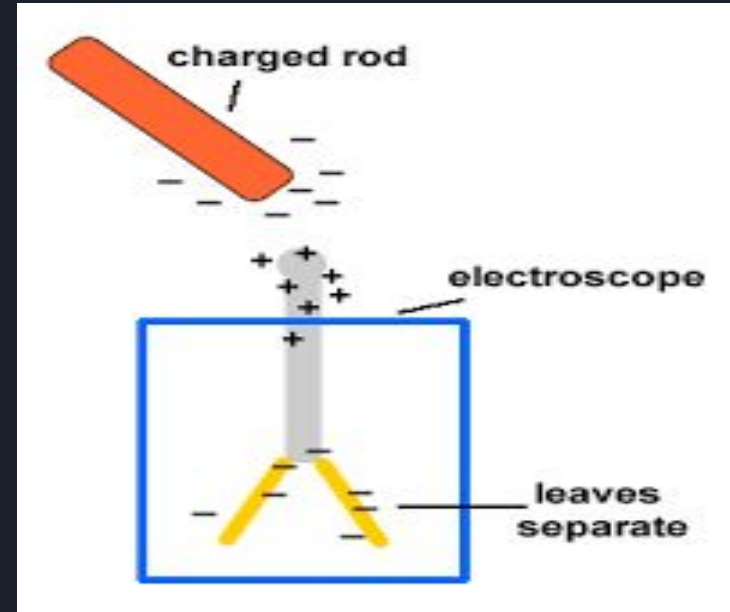
<https://circuitglobe.com/difference-between-conduction-and-induction.html>

Types of Static Charging: Type 3 - Induction

Induction occurs when a charged object is brought near a neutral object

INDUCTION CAUSES THE CHARGES ON THE ELECTROSCOPE TO POLARIZE

Since there is no contact between the rod and electroscope no electrons will move between the objects



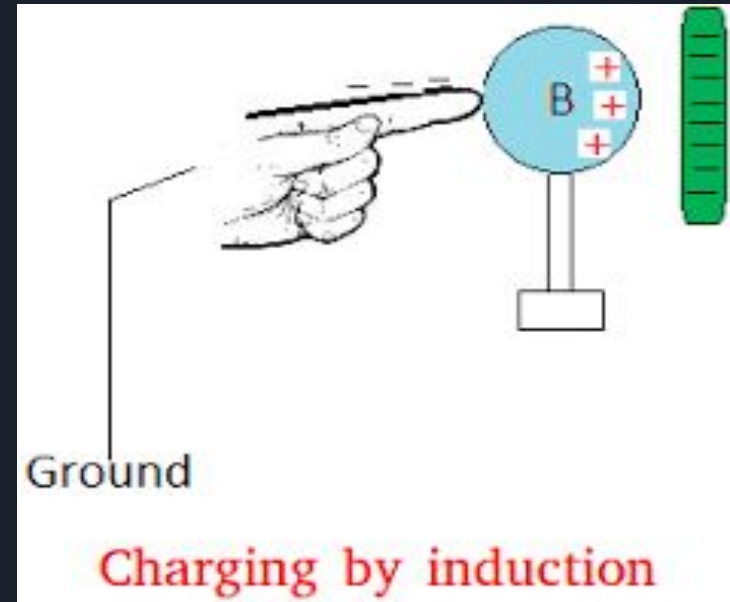
Types of Static Charging: Type 4 - Permanent Induction (Induction + Grounding)

Permanent induction occurs when a charged object, such as a rod, is brought near a neutral object, such as an electroscope, at the same time that the electroscope is grounded

INDUCTION CAUSES THE CHARGES ON THE ELECTROSCOPE TO POLARIZE; GROUNDING ALLOWS FOR ELECTRONS TO TRANSFER

Treat the rod and electroscope as an induction system and take the charge of the system

Electrons must transfer from the system to the grounding object (often your hand placed on the electroscope)



Source:
<https://www.basic-mathematics.com/charging-by-i>

Static Discharge

Static discharge is the sudden flow of electrons from an object or area that has a buildup of charges

Lighting and the Van De Graff Generator are examples of static discharge



Source:

<https://physicsworld.com/a/increased-lightning-hazard-almost-guarantee/>

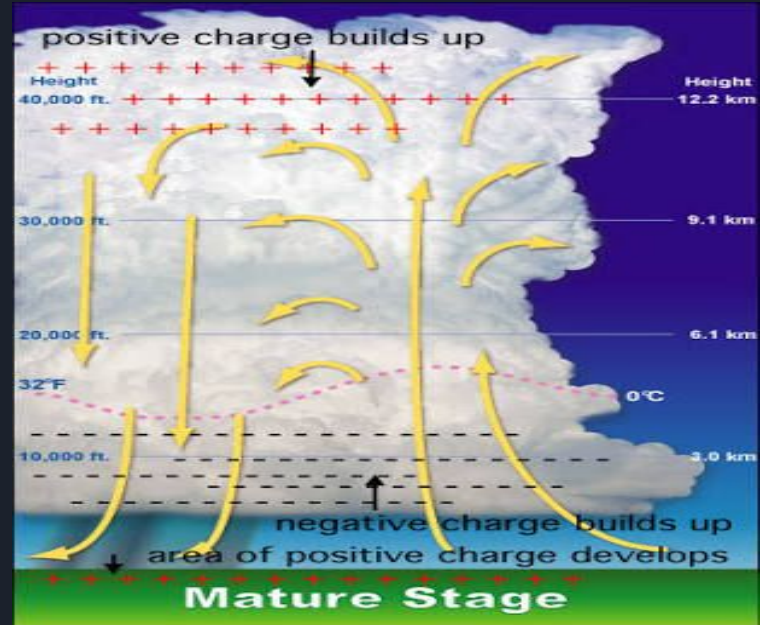


Lightning: Charging by Friction and Polarization

- Inside a storm cloud there are **updrafts** and **downdrafts** working together to create a washing machine effect
- **Updrafts** - warm air rising; light particles moving to the top of the cloud
- **Downdrafts** - heavier particles falling to the bottom of the cloud
- During the washing machine effect, the heavier and lighter particles rub against each other and are charged by friction
- During the collision, electrons are rubbed off the lighter ice crystals and given to the heavier particles making them negative
- The lighter particles at the top are now positive since they lost electrons

How lightning forms

- The massive build up of negative electrons at the base of the cloud **repels** the electrons on the ground
- Positive charges on the ground are **attracted** to the negative electrons from the cloud
- This **force** of attraction becomes so great that the electricity jumps the gap creating a lightning bolt





Static Discharge: The Van De Graff Generator

A Van De Graff generator is a machine that uses current electricity in order to produce **static electricity**

The purpose of the Van De Graff generator is to build up a lot of static charge

The Van De Graff generator has a metal brush at the top and bottom, a roller at the top and bottom made of different materials, a rubber belt, a motor, and a grounding wand (connected to a ground wire)

There are separate notes posted about the generator on Google Classroom as a Google Doc